

Lassonde School of Engineering

EECS2031

Practice Questions

Problem 1

Write a bash script for a simple calculator (save as `problem_1.sh`). The calculator can perform addition, subtraction, multiplication, division, min and max. The script **reads** the input from the standard input and displays the output on the standard output. The input is in one of 2 formats.

- 1) integer operation integer, where operation is one of + - * /. For example if the input is `6 * 7` the script should display `42` followed by a new line, nothing else
- 2) operation integer integer, where operation is either min or max. For example if the input is on the form `min 7 12`. Your script should display `12` followed by a new line.

Problem 2

Write a bash script that takes 2 integers as command line arguments (parameters) and prints the sum of the sequence of numbers between them

for example

bash problem_2.sh 4 8

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The output ends with a new line

Problem 3

Write a bash script (problem_3.sh) that does the following.

The script checks a file called "course_DB.txt". The file consists of records, each record is on a separate line. Each record consists of fields separated by tabs. The fields are

- Student name
- Student ID
- A variable number of fields with courses the student registered in
- The student GPA with 2 decimal places

For example (these are two records in the file)

```
John Doe 123456 EECS2032 EECS3201 6.75
```

```
Jane Doe 654321 EECS2011 EECS2020 EECS3201 EECS2121 5.21
```

The script runs as

problem_3.sh flag optional parameters

- If the flag is "-s" followed by student ID, the program prints the name of the student in a line by itself, and then in the next line it prints the courses the student is registered in separated by one space. If the ID is not in the file, it prints "No such ID". Then quits
- If the flag is "-m" The script prints the names and ID's of any student who is registered in more than 3 courses, and his/her GPA is less than 6.0. Then quits
- If the flag is "-c" followed by a number (floating point number), the script prints the names of the students whose GPA is less than the cutoff. Each name is in a line by itself. Then Quits

for example:

```
bash problem_3.sh -s 123456 John Doe EECS2032 EECS3201
```

```
bash problem_3.sh -m Jane Doe 654321
```

```
bash problem_3.sh -c 6.1 Jane Doe
```

Problem 4

Write sed commands in a file that does the following:

Any line that does not contain the pattern “ABABAB” must not be displayed (deleted). if the line that was not deleted, contains the word rain, change that to heavy rain

Save as problem_4.sed

Problem 5

Write a bash script that reads from a text file (p2a.txt).

Each line in the file contains either 4 or 5 integers.

Display the line where the first integer is greater than the sum of the second and third integers.

Save as problem_5.sh